02b — Repo Hosting: GitHub / GitLab / Bitbucket

Key Concepts

Term	Definition
Pull Request (PR)	GitHub term — proposed change from a branch, requires review before merge
Merge Request (MR)	GitLab term — same concept as PR
Branch Protection	Rules on `main` — require PR approval, passing CI, no direct push
GitHub Actions	GitHub's YAML-based CI/CD pipeline system
GitLab CI	GitLab's built-in CI/CD — no third-party tool needed
Fork	Copy of someone else's repo under your account (open source flow)
Clone	Local copy of a repo you have access to
Webhook	Notification sent to external services on repo events (push, PR, merge)

Platform Comparison

Feature	GitHub	GitLab	Bitbucket
Owned by	Microsoft	GitLab Inc	Atlassian
CI/CD	GitHub Actions	Built-in (GitLab CI)	Bitbucket Pipelines
Security scanning	Third-party	Built-in	Third-party
Container registry	Yes	Built-in	Limited
Self-hosted	GitHub Enterprise	Yes (free tier)	Yes
Best for	OSS, startups, big tech	Enterprises, compliance, DevOps	Atlassian/Jira shops

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Pull Request Flow (Industry Standard)

```
feature/my-feature
       |
       | git push
       v
  Remote Repo
       |
       | Open PR / MR
       v
  Automated Pipeline (YAML)
  ├── Linting
  ├── Unit tests
```

```
    ├── Integration tests
    └── Security scan
          |
          | All checks pass
          v
    Code Review (teammate approves)
          |
          | Merge
          v
      main branch
          |
          | CI/CD triggers deploy
          v
     Production
```

Why PRs Instead of Direct Pushes

- No broken code reaches `main` — automated checks catch it first
- Forces code review — second pair of eyes catches bugs and design issues
- Audit trail — every change has a linked PR with discussion and context
- Branch protection enforces this at the platform level — can't bypass it

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Fork vs Clone

	Fork	**Clone**
Used for	Contributing to someone else's repo	Working on a repo you have access to
Creates copy	Under your GitHub account	On your local machine
Flow	Fork → clone fork → PR to original	Clone → branch → PR to same repo

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Industry Examples

Company	**Usage**
Microsoft	Hosts VS Code, TypeScript, .NET on GitHub — 100M+ developers
GitLab	Chosen by enterprises needing self-hosted, all-in-one DevOps platform
Atlassian shops	Bitbucket + Jira = issue-to-PR traceability out of the box

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Interview Questions

Q: Why would a company choose GitLab over GitHub? A: GitLab bundles CI/CD, security scanning, container registry, and issue tracking in one platform. Fewer vendors, self-hostable for compliance, no stitching together separate tools.

Q: What is a Pull Request and why does every team use them? A: A PR is a proposed merge that triggers automated checks and requires peer review before code reaches main. It prevents broken or unreviewed code from reaching production.

Q: What is branch protection? A: Rules enforced at the platform level on protected branches (usually `main`) — require PR approval, passing CI, no direct pushes. Ensures quality gates can't be bypassed.

Q: What's the difference between a fork and a clone? A: Clone copies a repo to your local machine. Fork copies it to your own remote account — used in open source so you can PR back to the original without write access.